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Practice Lab Assignments: 1

1. To accept an object mass in kilograms and velocity in meters per second and display its momentum. Momentum is calculated as momentum= mass*velocity where m is the mass of the object and v is its velocity.

Code:

```
m = float(input("Enter mass(kg): "))

v = float(input("Enter velocity(m/s): "))

momentum = m * v

print("Momentum = ",momentum, "kg.m/s")
```

---

2. Write a Python program for following conditions.

- If n is single digit print square of it.
- If n is two digit print square root of it.
- If n is three digit print cube root of it.

Code:

```
import math

n = int(input("Enter a number: "))

if 0 < n < 10:

    square = n * n

    print("Square = ", square)

elif 10 < n < 100:

    square_root = n ** 0.5

    print("Square root = ", square_root)

elif 100 < n < 1000:

    cube_root = n ** (1/3)

    print("Cube root = ", cube_root)

else:

    print("Please enter a number between 0 and 999.")
```

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3. Read the birthdate and salary in rupees of employees. Perform data transformation for birthdate to age and also salary which is in rupees to salary in dollars using functions.

Code: from datetime import datetime

```
def calculate_age(birthdate):
    today = datetime.now()
    birthdate = datetime.strptime(birthdate, "%Y-%m-%d")
    return today.year - birthdate.year - ((today.month, today.day) < (birthdate.month, birthdate.day))
def salary_in_dollars(salary):
    return salary /82.5


birthdate = input("Enter birthdate (YYYY-MM-DD): ")
salary = int(input("Enter salary in rupees: "))


age = calculate_age(birthdate)
salary_usd = salary_in_dollars(salary)


print("Age:", age)
print("Salary in USD: ", round(salary_usd, 2))
```

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4. Print the reverse number of a given number.

Code:

```
num = input("Enter a number: ")
print("Reverse number is:", num[::-1])
```

Lab Assignment: 1

1. To accept students five courses marks and compute his/her result. Student is passing if he/she scores marks equal to and above 40 in each course. If student scores aggregate greater than 75 percentage, then the grade is distinction.

- If aggregate is greater than or equal to 60 and less than 75 then the grade if first division.
- If aggregate is greater than or equal 50 and less than 60, then the grade is second division.
- If aggregate is greater than or equal 40 and less than 50, then the grade is third division.

Code:

```
marks = [int(input(f"Enter marks {i}: ")) for i in range(1, 6)]

if min(marks) < 40:

    print("Fail")

else:

    per = sum(marks) / 500 * 100

    print("Percentage:", per)

    if per > 75:

        print("Distinction")

    elif per >= 60:

        print("First Division")

    elif per >= 50:

        print("Second Division")

    else:

        print("Third Division")
```

---

2. Solve the Fibonacci sequence using recursive function in Python.

Code:

```
def fib(n):

    if n <= 1:

        return n

    return fib(n-1) + fib(n-2)
```

```
    n = int(input("Enter number: "))
print(fib(n))
```

3. Write a Python program to print different patterns.

Code:

```
n = int(input("Enter number: "))
for i in range(1, n+1):
    print("*" * i)


n = int(input("Enter number: "))
for i in range(n, 0, -1):
    print("*" * i)
```

Self Study Assignment: 1

Perform all statistical analysis (Average, Max, Min, Sum) on employee data.

Code:

```
n = int(input("Enter number of employees: "))
emp = []
for i in range(n):
    emp.append(int(input("Enter salary: ")))
print("Sum =", sum(emp))
print("Average =", sum(emp)/n)
print("Max =", max(emp))
print("Min =", min(emp))
```

Practice Lab Assignment: 2

1. Perform all List Operation, Dictionary Operations.

Code:

```
student={"name":"Reshma Kumare","age":18,"grades":[85,90,98]}

print("Name:",student["name"])
```

Name: Reshma Kumare

```
student["PRN"] = 202501080017

print("After adding PRN:",student)
```

After adding PRN: {'name': 'Reshma Kumare', 'age': 18, 'grades': [85, 90, 98], 'PRN': 202501080017}

```
student["age"] = 19

print("After updating age:",student)
```

After updating age: {'name': 'Reshma Kumare', 'age': 19, 'grades': [85, 90, 98], 'PRN': 202501080017}

```
student.pop("grades")
```

[85, 90, 98]

```
print("After removing grades:",student)
```

After removing grades: {'name': 'Reshma Kumare', 'age': 19, 'PRN': 202501080017}

```
keys = student.keys()

print("Keys:", keys)
```

Keys: dict_keys(['name', 'age', 'PRN'])

```
values = student.values()

print("Values:",values)
```

Values: dict_values(['Reshma Kumare', 19, 202501080017])

```
student_copy = student.copy()
print("Copy of dictionary:", student_copy)
```

Copy of dictionary: {'name': 'Reshma Kumare', 'age': 19, 'PRN': 202501080017}

```
student.clear()
print("After clear:", student)
```

After clear: {}

Lab Assignment: 2

1. Select the number from the entered list and find its position in Python (use Linear Search).

Code:

```
lst = []
n = int(input("Enter number of elements: "))
for i in range(n):
    lst.append(int(input("Enter number: ")))
x = int(input("Enter number to search: "))
for i in range(n):
    if lst[i] == x:
        print("Position:", i+1)
        break
else:
    print("Number not found")
```

---

2. Choose cricket team of eleven players find the captain of the team (consider tallest person as a captain) using dictionary.

Code:

```
team = {}
for i in range(11):
    name = input("Enter player name: ")
    height = int(input("Enter height: "))
    team[name] = height
```

```
max_height = 0
captain = ""
for name in team:
    if team[name] > max_height:
        max_height = team[name]
        captain = name
print("Captain of the team is:", captain)
```

Self Study Assignment: 2

1. Consider the student subjects marks are stored in the list. Find the total marks and percentage of students.

Code:

```
marks = []
n = int(input("Enter number of subjects: "))
for i in range(n):
    m = int(input("Enter marks: "))
    marks.append(m)
total = sum(marks)
percentage = total / n
print("Total Marks:", total)
print("Percentage:", percentage, "%")
```

---

2. Perform all tuple operations.

Code:

```
t = (10, 20, 30, 40, 50)
print("Tuple:", t)
```

Tuple: (10, 20, 30, 40, 50)

```
print("Length:", len(t))
```

Length: 5

```
print("First element:", t[0])
```

First element: 10

```
print("Slice (1 to 3):", t[1:4])
```

Slice (1 to 3): (20, 30, 40)

```
print("Count of 20:", t.count(20))
```

Count of 20: 1

```
print("Index of 30:" t.index(30))
```

SyntaxError: invalid syntax. Perhaps you forgot a comma?

```
print("Index of 30:", t.index(30))
```

Index of 30: 2

```
t2 = (60, 70)
print("After adding:", t + t2)
```

After adding: (10, 20, 30, 40, 50, 60, 70)